

FOR SERVICE TECHNICIAN'S USE ONLY

DANGER



Electrical Shock Hazard

Only authorized technicians should perform diagnostic voltage measurements.

After performing voltage measurements, disconnect power before servicing.

Failure to follow these instructions can result in death or electrical shock.

WARNING



Electrical Shock Hazard

Disconnect power before servicing.

Replace all parts and panels before operating.

Failure to do so can result in death or electrical shock.

Voltage Measurement Safety Information

When performing live voltage measurements, you must do the following:

- Verify the controls are in the off position so that the appliance does not start when energized.
- Allow enough space to perform the voltage measurements without obstructions.
- Keep other people a safe distance away from the appliance to prevent potential injury.
- Always use the proper testing equipment.
- After voltage measurements, always disconnect power before servicing.

IMPORTANT: Electrostatic Discharge (ESD) Sensitive Electronics

ESD problems are present everywhere. Most people begin to feel an ESD discharge at approximately 3000V. It takes as little as 10V to destroy, damage, or weaken the main control assembly. The new main control assembly may appear to work well after repair is finished, but a malfunction may occur at a later date due to ESD stress.

- Use an anti-static wrist strap. Connect wrist strap to green ground connection point or unpainted metal in the appliance

-OR-

Touch your finger repeatedly to a green ground connection point or unpainted metal in the appliance.

- Before removing the part from its package, touch the anti-static bag to a green ground connection point or unpainted metal in the appliance.
- Avoid touching electronic parts or terminal contacts; handle electronic control assembly by edges only.
- When repackaging main control assembly in anti-static bag, observe above instructions.

IMPORTANT SAFETY NOTICE — “For Technicians only”

This service data sheet is intended for use by persons having electrical, electronic, and mechanical experience and knowledge at a level generally considered acceptable in the appliance repair trade. Any attempt to repair a major appliance may result in personal injury and property damage. The manufacturer or seller cannot be responsible, nor assume any liability for injury or damage of any kind arising from the use of this data sheet.

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WHIRLPOOL CONTROL PANEL (features and appearances may vary between models)

Rotating the cycle selector knob turns off each corresponding cycle indicator. (Features and appearances vary between models).

Color LCD Screen – pressing the buttons around the screen changes their on-screen representations from black to green. UI Component Test, Error Codes, Diagnostics, and System Information are accessed via Service Menu.

“POWER” button: press once to turn off indicator.

“START/PAUSE” button: press once to turn off indicator. Begin or continue test.

“Static Reduce” button:
press once to turn off indicator
(not available on all models).

“EcoBoost” button: press once to turn off indicator (not available on all models).

“Back & (◀)” buttons: press one or more times to exit Diagnostic Mode.

(1st, 2nd, 3rd) Recommended buttons to access Diagnostic Mode.

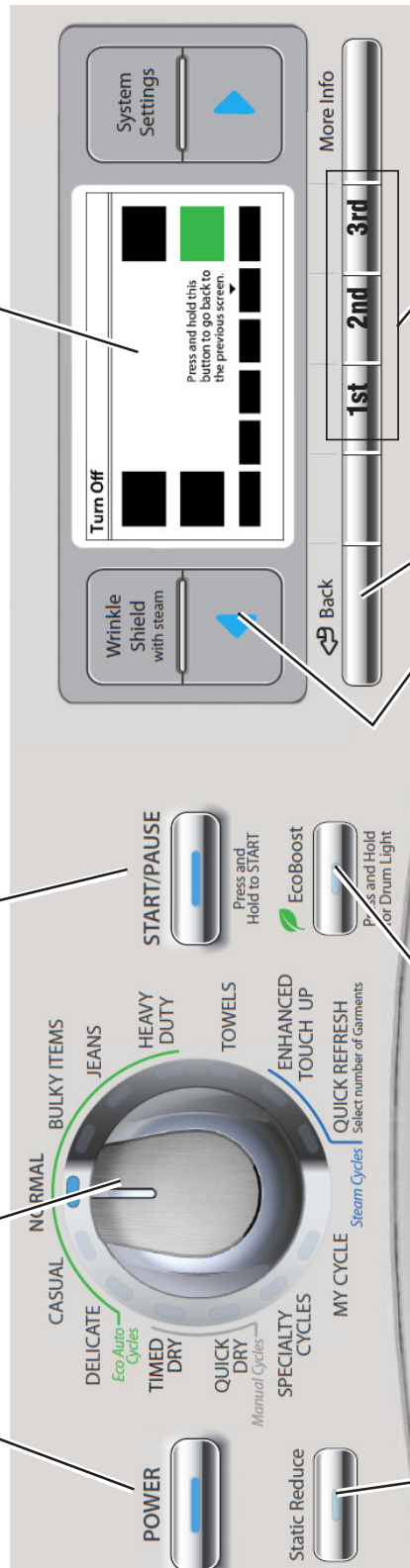


Figure 1 - UI Component Test

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DIAGNOSTIC GUIDE

Before servicing, check the following:

- Make sure there is power at the wall outlet.
- Has a household fuse blown or circuit breaker tripped? Was a regular fuse used? Inform customer that a time-delay fuse is required.
- Is dryer vent properly installed and clear of lint or obstructions?
- All tests/checks should be made with a VOM (volt-ohm-milliammeter) or DVM (digital-voltmeter) having a sensitivity of 20,000 Ω per volt DC or greater.
- Resistance checks must be made with dryer unplugged or power disconnected.
- **IMPORTANT:** Avoid using large diameter probes when checking harness connectors as the probes may damage the connectors upon insertion.
- Check all harnesses and connections before replacing components. Look for connectors not fully seated, broken or loose wires and terminals, pin insertion, or wires not pressed into connectors far enough to engage metal barbs.
- A potential cause of a control not functioning is corrosion or contamination on connections. Use an ohmmeter to check for continuity across suspected connections.

SERVICE DIAGNOSTICS

These tests allow factory or service personnel to test and verify all inputs to the machine control electronics. You may want to do a quick and overall checkup of the dryer with these tests before going to specific troubleshooting tests.

ACTIVATING SERVICE DIAGNOSTICS

1. Be sure the dryer is in standby mode (plugged in with all indicators off).
2. Select any three buttons (except POWER) and follow the steps below, using the same buttons. Remember the buttons and the order that the buttons were pressed. (Recommended button selections shown in figure 1):

Within 8 seconds,

- Press and Release the **1st** selected button,
 - Press and Release the **2nd** selected button,
 - Press and Release the **3rd** selected button;
 - Repeat this 3 button sequence 2 more times.
3. If the key sequence was entered successfully, the language selection screen is displayed.
 4. Select the desired language—choices are: English, French, and Spanish.
 5. The following screen warns that this area is for service technicians only.
 - Pressing “Exit” returns to original screen.
 - Pressing “Enter” continues to DIAGNOSTIC HOME screen.

NOTE: Service Diagnostics will time out and return to standby after 5 minutes of user inactivity.

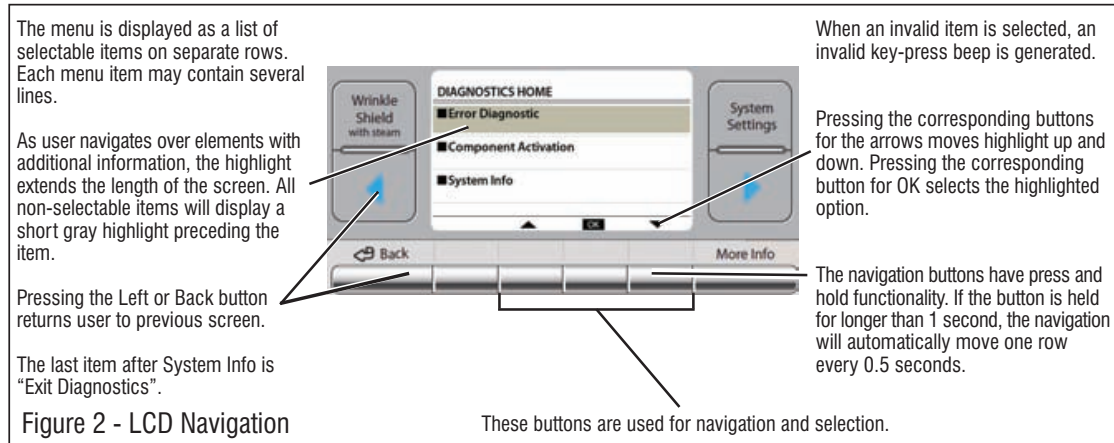
EXITING SERVICE DIAGNOSTICS

Use either of the two methods below to exit Service Diagnostics.

- Select “Exit Diagnostics” on the Diagnostics Home Screen.
- Press the “Back” button one or more times, depending on location in Service Diagnostics.

DIAGNOSTICS HOME SCREEN

LCD Navigation (See figure 2)



DIAGNOSTICS HOME SCREEN

The "Diagnostics Home" screen (see figure 2, page 3) displays the following service diagnostic items:

- **Error Diagnostic** – Access Error Codes, Error History, Quick Diagnostic Test, and Clear All Errors
- **Component Activation** – Activation and functional testing of system components and sensors (page 5)
- **System Info** – Displays model #, serial #, hardware info, and other system variables (page 5)
- **Exit Diagnostics** – Exits Service Diagnostics and returns to standby mode

ERROR DIAGNOSTIC

Access Error Codes, Error History, Quick Diagnostics, and Clear All Errors

NOTE: If no saved error exists, the only menu item displayed is "Quick Diagnostic Test".

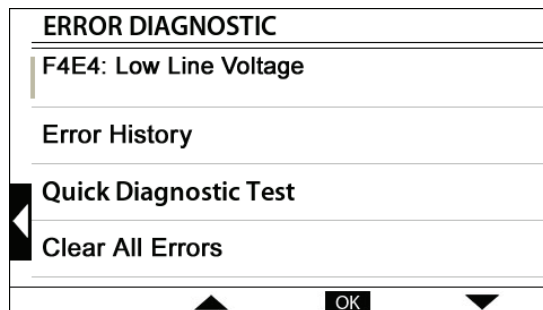


Figure 3 - Error Diagnostic Screen

- **Error Codes** – Displays most recent saved fault/error code (if no saved error exists, this item is NOT shown). Refer to Fault/Error Code chart on page 7.

- **Error History** – Displays the last four saved fault/error codes (if no saved error exists, this item is NOT shown). Provides error code and description, time of error, and cycle information when error occurred (see figure 4).

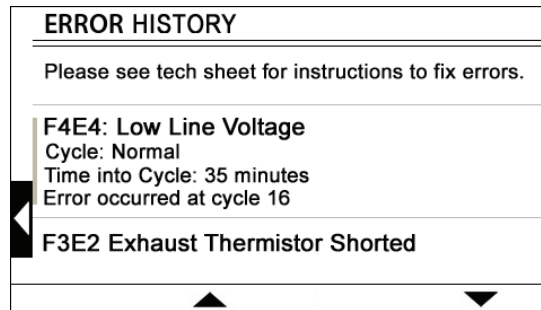


Figure 4 - Error History Screen

- **Quick Diagnostic Test** – Performs a quick system level test of the dryer and displays error results in a dialog box (see figure 5). Tests drum motor and heater(s), exhaust and inlet thermistors, air flow (AF), and low line voltage (L2). Test stops on first error detected. Use this test to check for errors and verify repairs. **NOTE:** Door must be closed to perform test.

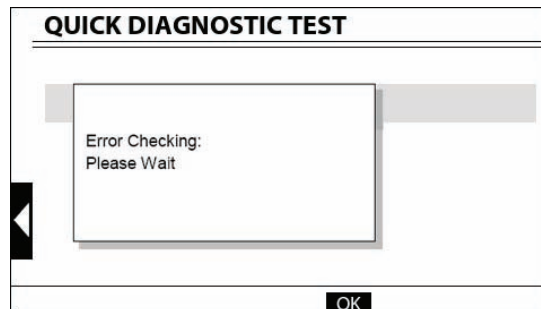


Figure 5 - Quick Diagnostic Test

- **Clear All Errors** – Clears all stored fault/error codes (if no saved error exists, this item is NOT shown). **NOTE:** Once the errors are cleared, they cannot be recovered again.

COMPONENT ACTIVATION

Activation and functional testing of system components and sensors (see Figure 6).

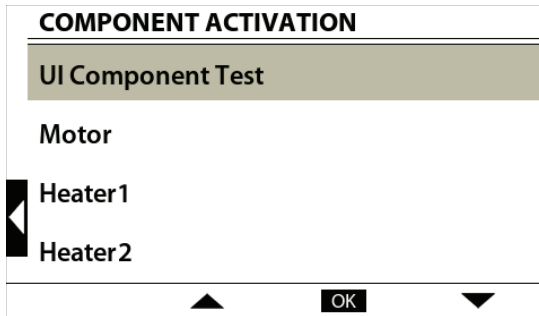


Figure 6 - Component Activation Menu

- **Component Activation** – When a component is selected from the Component Activation menu (see Figure 6), a screen appears providing component information, and if available, attributes related to the component (see figure 7). Many of the components listed can be turned on and off. By default, the component is off. (See chart on page 6.)

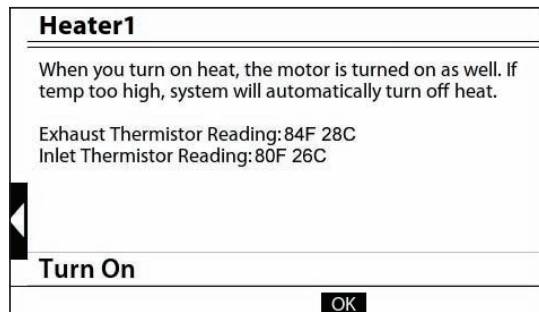


Figure 7 - Component Screen

- **UI Component Test** – Upon activation, all LEDs are turned on. Pressing each button turns off its LED, rotating the cycle selector knob turns off each LED, and pressing the ATM buttons changes their on-screen representations from black to green (see figure 8). Press and hold the right-most navigation key to return to previous screen.

- If indicators do not turn off and beep after pressing buttons and rotating the cycle selector, go to TEST #6: Buttons & Indicators, page 19.

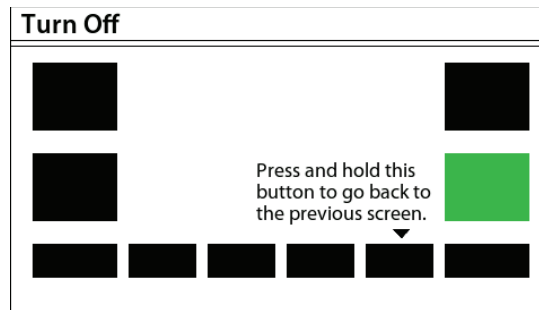


Figure 8 - UI Component Test

SYSTEM INFO

Displays model and serial number, hardware information, and other system variables (if provided).

NOTE: Missing system information does not indicate a problem. The system information screens are common across various platforms; therefore, information provided on one dryer may not be available on another. If uncertain about the information, check with technical support.

NOTE: All non-selectable items will display a short gray highlight preceding the selection; however, as the highlight of the screen moves over items with more information, it displays regular background length. (see Figures 9 & 10.)

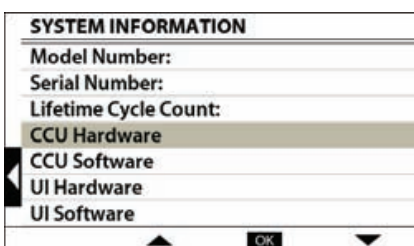


Figure 9 - System Information Screen

Non-selectable items →
← Items with more info.

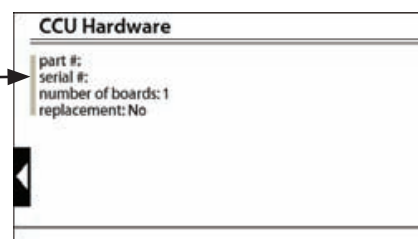


Figure 10 - Hardware Information Details

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Component Activation Chart

COMPONENT ACTIVATION CHART		
Output	On/Off	Details
UI Component Test	N/A	Upon activation, all LEDs are turned on. Pressing each button turns off its LED. Rotating the cycle selector knob turns off each LED. Pressing the ATM buttons changes their on-screen representations from black to green.
Motor	YES	Status (On/Off) NOTE: Door must be closed. If the motor does not turn on, go to TEST #3, page 12.
Heater1 (Electric)/ Heat On (Gas)	YES	Status (On/Off) NOTE: Door must be closed. Attributes: Exhaust & Inlet Thermistor Reading. When you turn on heat, the motor is turned on as well. If temp is too high, system will automatically turn off heat. • If the motor does not turn on, go to TEST #3, page 12. • If no heat is detected, go to TEST #4, page 13.
Heater2 (Electric Only)	YES	Status (On/Off) NOTE: Door must be closed. Attributes: Exhaust & Inlet Thermistor Reading. When you turn on heat, the motor is turned on as well. If temp is too high, system will automatically turn off heat. • If the motor does not turn on, go to TEST #3, page 12. • If no heat is detected, go to TEST #4, page 13.
Heater 1 & 2 (Electric Only)	YES	Status (On/Off) NOTE: Door must be closed. Attributes: Exhaust & Inlet Thermistor Reading. When you turn on heat, the motor is turned on as well. If temp is too high, system will automatically turn off heat. • If the motor does not turn on, go to TEST #3, page 12. • If no heat is detected, go to TEST #4, page 13.
Inlet Thermistor	N/A	Status (F°, C°, and Resistance) If inlet thermistor reading is out of range, go to TEST #4a, page 16.
Exhaust Thermistor	N/A	Status (F°, C°, and Resistance) If exhaust thermistor reading is out of range, go to TEST #4a, page 16.
Moisture Sensor	N/A	Status (Open/Closed) • Place finger over both sensor strips. If sensor status does not change, go to TEST #5, page 18.
Door Status	N/A	Status (Open/Closed) • If opening the door does not change status, go to TEST #7, page 20.
CCU Line Voltage 1	N/A	Status (L1 Voltage Measurement) NOTE: Door must be closed. When measuring L1, the motor and heater are activated. • If L1 is absent, go to TEST #1, page 9.
CCU Line Voltage 2 (Electric Only)	N/A	Status (L2 Voltage Measurement) NOTE: Door must be closed. When measuring L2, the motor and heater are activated. • If L2 is absent, go to TEST #1, page 9.
Air Flow	N/A	Status (Normal/Restricted) NOTE: Door must be closed. • If air flow does not turn on, go to TEST #4, page 13.
Drum Light	YES	Status (On/Off) • If drum light does not turn on, go to TEST #1, page 9.
Buzzer	YES	Status (On/Off) • If buzzer does not sound, go to TEST #6, page 19.
Myst Valve	YES	Status (On/Off) After being turned on, the myst valve automatically deactivates after 30 seconds if not turned off. • If myst valve does not turn on, go to TEST #9, page 21.

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Fault/Error Code Chart

The fault codes below may be indicated under various conditions and can be accessed through Service Diagnostic Mode.

CODE	DESCRIPTION	EXPLANATION AND RECOMMENDED PROCEDURE
CUSTOMER FAULT/ERROR CODES		
	Power Failure / Interruption	Indicates that a power failure occurred while the dryer was running. Press START to continue the cycle, or press POWER to clear the display.
	Restricted Air Flow	Indicates low air flow that may affect dryer performance. Confirm that airflow system is not blocked. Check lint screen, exhaust duct, exhaust fan. • Refer to Fault/Error Code "F4E3" below for recommended procedure.
	Low Line Voltage	Indicates low L2 voltage (less than 30 V) is detected at the CCU. • Refer to Fault/Error Code "F4E4" below for recommended procedure.
SERVICE FAULT/ERROR CODES		
F1E1	Cycle Control Unit (CCU) Problem	Indicates a CCU problem. • Verify all connections to and from the CCU. • See TEST #1: CCU Power Check, page 9.
F2E1	User Interface (UI) Problem (stuck button)	Indicates a stuck button (depressed for over 20 seconds) or UI mismatch. This fault code will ONLY appear in the service diagnostic mode. See TEST #6: Buttons and Indicators, page 19.
F2E2	User Interface Software Error 1	Appears if UI cannot read EEPROM software from UI. • Verify all connections between CCU and UI. • Replace the UI.
F2E3	User Interface Software Error 2	Appears if main control EEPROM is not programmed correctly. • Replace the UI.
F3E1	Exhaust Thermistor Open	Indicates that the exhaust thermistor is open. Temperature drops below 18° F (> 50k ohms). See TEST #4a: Thermistors, page 16.
F3E2	Exhaust Thermistor Shorted	Indicates that the exhaust thermistor has shorted. Temperature above 250° F (< 500 ohms). See TEST #4a: Thermistors, page 16.
F3E3	Inlet Thermistor Open	Indicates that the Inlet Thermistor is open. Temperature drops below 18° F (>245k ohms). See TEST #4a: Thermistors, page 16.
F3E4	Inlet Thermistor Shorted	Indicates that the Inlet Thermistor is shorted. Temperature above 391° F (< 328 ohms). See TEST #4a: Thermistors, page 16.
F3E5	Inlet and Exhaust Thermistor Open	Indicates that the Inlet and Exhaust Thermistor are open. Will occur if the P14 connector is not plugged into the CCU.
F3E6	Moisture Sensor Open	Indicates that the moisture sensor strip is open. This fault code will ONLY appear in the service diagnostic mode. See TEST #5: Moisture Sensor, page 18.
F3E7	Moisture Sensor Shorted	Indicates that the moisture sensor strip has shorted. This fault code will ONLY appear in the service diagnostic mode. See TEST #5: Moisture Sensor, page 18.
F4E1	Heater Relay or Connector Problem	Indicates no voltage detected at the heater relay. This fault code will ONLY appear in the service diagnostic mode. • Unplug dryer or disconnect power and check that the wires are plugged into the heater element(s) and the relay(s) on the CCU.
F4E3	Restricted Air Flow	Indicates low air flow that may affect dryer performance. • Confirm that airflow system is not blocked, check lint screen, exhaust duct, exhaust fan. • See TEST #4a: Thermistors, page 16.
F4E4	L2 Line Voltage Error	L2 indicates low L2 voltage (less than 30 V) is detected at the CCU. • Check to see if a household fuse has blown or a circuit breaker has tripped. • Confirm the power cord is properly installed and plugged into the power outlet. • Check the relay connections on the CCU. • Gas Models Only: Check the P14 connection on the CCU. (Harness Loopback on pins 4 & 5)
F6E1 F6E2	Communication Error UI to CCU Communication Error CCU to UI	Communication between the CCU and UI has not been detected. • Check the harness continuity and connections between the CCU and UI. • Check AC and DC supplies. See TEST #1: CCU Power Check, page 9. • Replace the User Interface. • Replace the CCU.

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TROUBLESHOOTING GUIDE

Some tests will require accessing components (see page 27).

PROBLEM	POSSIBLE CAUSE	CHECKS & TESTS
WILL NOT POWER UP - No operation - No keypad response - No LEDs or display	No power to dryer.	Check power at outlet, check circuit breaker, fuses, or junction box connections.
	Connection problem between AC plug and dryer.	See Test #2: Supply Connections, page 10.
	Connection problem between CCU and UI.	Check connections and continuity between CCU and UI.
	Power supplies not present at machine electronics.	See Test #1: CCU Power Check, page 9.
	User Interface problem.	See Test #6: Buttons & Indicators, page 19.
WILL NOT START CYCLE (No response when Start button is pressed.)	Door not fully closed or striking the door latch.	Be sure the door is completely closed, then press and hold the START button.
	Door Switch problem.	See Test #7: Door Switch, page 20.
	Drive Belt / Belt Switch problem.	See Test #3: Motor Circuit, page 12.
	Thermal Fuse / Motor problem.	See Test #3: Motor Circuit, page 12.
	User Interface problem.	See Test #6: Buttons & Indicators, page 19.
WILL NOT SHUT OFF WHEN EXPECTED	CCU problem.	See Test #1: CCU Power Check, page 9.
	Poor airflow.	Check lint screen and exhaust vent. Clean if necessary.
	Check the Pause/Cancel button.	Perform UI Component Test under Component Activation.
	Moisture Sensor problem.	See Test #5: Moisture Sensor, page 18.
	Thermistor problem.	See Test #4a: Thermistors, page 16.
CONSOLE WON'T ACCEPT SELECTIONS	User Interface problem.	See Test #6: Buttons & Indicators, page 19.
	User selects invalid option.	Refer customer to "Use and Care Guide".
DRUM WILL NOT SPIN	CCU problem.	See Test #1: CCU Power Check, page 9.
	Motor problem.	See Test #3: Motor Circuit, page 12.
	Door switch problem.	See Test #7: Door Switch, page 20.
	Thermal Fuse (elect. Only).	See Test #4b: Thermal Fuse, page 17.
	Drive Belt / Belt Switch problem.	See Test #3: Motor Circuit, page 12.
WILL NOT HEAT	CCU problem.	See Test #1: CCU Power Check, page 9.
	Heater system malfunction or open heater coil.	See Test #4: Heat System, page 13.
	Check for L1 and L2.	Perform CCU, L1, and L2 tests in Service Diagnostics.
	Check installation.	Verify proper dryer installation.
HEATS IN AIR CYCLE	CCU problem.	See Test #1: CCU Power Check, page 9.
	Heater system problem.	See Test #4: Heat System, page 13.
	Heater relay shorted.	See Test #4: Heat System, page 13.
SHUTS OFF BEFORE CLOTHES ARE DRY	Heater coil shorted.	See Test #4: Heat System, page 13.
	Dryness setting for auto cycles.	Increase drying times for one or more auto cycles.
	Lint screen full.	Clean if necessary. Refer customer to "Use and Care Guide".
	Heater vent clogged.	Clean if necessary. Refer customer to "Use and Care Guide".
	Moisture Sensor problem.	See Test #5: Moisture Sensor, page 18.
WATER VALVE NOT DISPENSING (Water valve is activated intermittently during the steam cycle.)	Adjust Customer Focused Dryness Level.	See Test #5a: Adjusting Customer-Focused Dryness Level, page 19.
	No water to valve.	Verify water supply is turned on.
	No water from valve.	See Test #9: Myster Valve, page 21.
	Steam cycle not selected.	Refer customer to "Use and Care Guide".

FOR SERVICE TECHNICIAN'S USE ONLY

TROUBLESHOOTING TESTS

IMPORTANT: The following procedures may require the use of needle probes to measure voltage. Failure to use needle probes will damage the connectors.

TEST #1: CCU Power Check

This test is used to determine if power is present at the machine control electronics. This test assumes that proper voltage is present at the outlet.

1. Unplug dryer or disconnect power.
2. Check for appropriate line voltages at the outlet: 240VAC (electric 2-phase), 208VAC (electric 3-phase), or 120VAC (gas).
- If line voltage is present, go to step 3.
- If line voltage is not present, check for tripped circuit breaker or blown household fuse. If CB (circuit breaker) is not tripped, have customer check with qualified electrician.
3. Remove top panel to access the machine electronics.
4. Plug in dryer or reconnect power.

5. CCU VAC – With voltmeter set to **AC**, connect black probe to CCU P8-3 (N) and red probe to P9-2 (L1). (See Figure 11.)

- If 120VAC is present, go to step 6.
- If 120VAC is not present, perform TEST #2: Supply Connections, page 10.

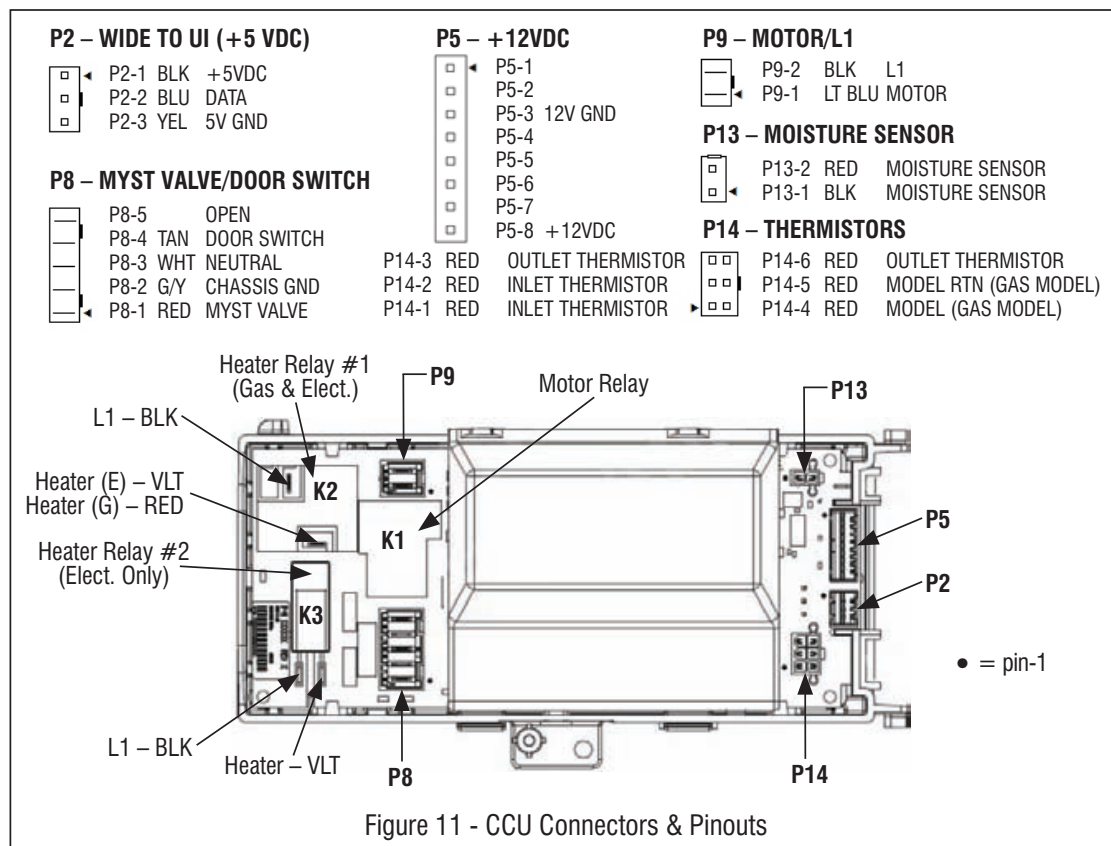
6. CCU +5VDC – With voltmeter set to **DC**, connect black probe to CCU P2-3 (ground) and red probe to P2-1 (+5V DC).

- If +5VDC is present, go to step 9.
- If +5VDC is not present, go to step 7.

7. Unplug dryer or disconnect power. Unplug P14 from the CCU. Plug in dryer or reconnect power and repeat step 6.

- If +5VDC returns, one of the thermistors has shorted. To diagnose thermistors, see TEST #4a, page 16.
- If +5VDC is not present, go to step 8.

8. Unplug dryer or disconnect power. Reconnect P14 to the CCU and unplug P2 from the CCU. Plug in dryer or reconnect power and repeat step 6. Perform voltage check inside header P2 on CCU, between pins 1 & 3—DO NOT SHORT PINS TOGETHER.



- If +5VDC is still missing, replace the CCU.
- If +5VDC returns, check harnesses and connections between the CCU and user interface (UI). If acceptable, replace the UI.

9. CCU +12VDC – with voltmeter set to **DC**, connect black probe to CCU P5-3 (ground) and red probe to P5-8 (+12VDC).

- If +12VDC is present, go to step 10.
 - If +12VDC is not present, replace the CCU.
- NOTE:** 12V is used to actuate the 120VAC relays, triacs, and switches.

10. Unplug dryer or disconnect power.

11. Reassemble all parts and panels.

12. Perform the “Quick Diagnostic Test” through Service Diagnostics to verify repair.

TEST #2: Supply Connections

This test assumes that proper voltage is present at the outlet, and for U.S. installations, a visual inspection indicates that the power cord is securely fastened to the terminal block (electric dryer) or wire harness connection (gas dryer).

ELECTRIC DRYER (U.S. Installations):

1. Unplug dryer or disconnect power.
2. Remove the cover plate from the top right corner of the back of the dryer. See figure 12.

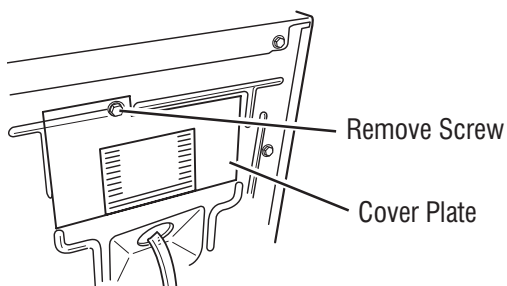


Figure 12 - Remove the cover plate.

3. With an ohmmeter, check for continuity between the neutral (N) terminal of the plug and the center contact on the terminal block. See figure 13a.
- If there is no continuity, replace the power cord and test the dryer.
 - If there is continuity, go to step 4.
4. In a similar way, check which terminal of the plug is connected to the left-most contact on the terminal block and make a note of it. This will be L1 (black wire) in the wiring diagram. See figure 13a.

- When this is found, go to step 5.
- If neither of the plug terminals have continuity with the left-most contact of the terminal block, replace the power cord and retest dryer.

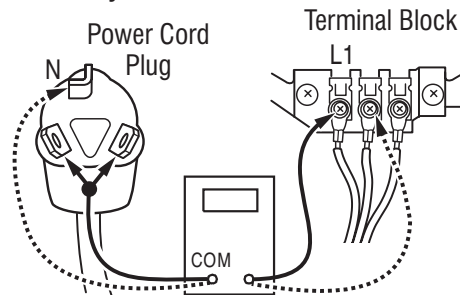


Figure 13a - Plug-to-terminal connections for electric dryer.

5. Access the machine electronics without disconnecting any wiring to the CCU (See Dryer Disassembly Instructions, page 23).
 6. With an ohmmeter, check for continuity between the L1 terminal of the plug (found in step 4) and P9-2 (black wire) on the CCU.
- If there is continuity, go to step 7.
 - If there is no continuity, check that wires to the terminal block are mechanically secure. If so, replace the main wire harness in the dryer and retest.
7. With an ohmmeter, check for continuity between the neutral (N) terminal of the plug and P8-3 (white wire) on the CCU.
- If there is continuity, go to step 8.
 - If there is no continuity, and the mechanical connections of the wire are secure, replace the main wire harness in the dryer and retest.
8. Visually check that ALL connectors are fully inserted into the CCU.
 9. Reassemble all parts and panels.
 10. Plug in dryer or reconnect power.
 11. Perform the “Quick Diagnostic Test” through Service Diagnostics to verify repair.

ELECTRIC DRYER (Canadian Installations):

1. Unplug dryer or disconnect power.
2. Remove the cover plate from the top right corner of the back of the dryer. See figure 12.

3. Access the machine electronics without disconnecting any wiring (see Dryer Disassembly Instructions, page 23).

4. With an ohmmeter, check the continuity from L1 and N plug terminals of the power cord to the terminals for L1 and N on the CCU. See figure 13b.

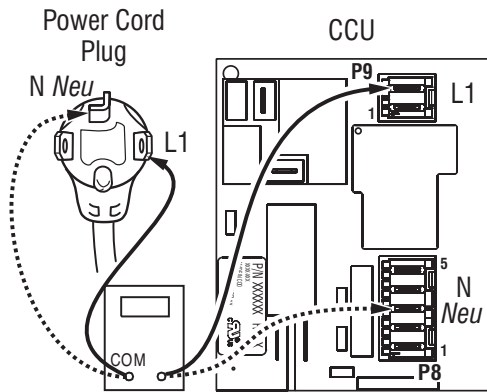


Figure 13b - Plug-to-terminal connections for electric dryer.

- If continuity exists for both connections, go to step 6.
- If an open circuit is found, check the integrity of the connections of the power cord to the harness in the dryer; harness to the CCU; and the integrity of the power cord itself.

5. If it is necessary to replace the power cord, remove the retaining clip that secures the cord to the back panel. Disconnect the cord from the main harness and the ground wire from the rear panel, then pull out the power cord.

6. Visually check that ALL connectors are fully inserted into the CCU.

7. Reassemble all parts and panels.

8. Plug in dryer or reconnect power.

9. Perform the "Quick Diagnostic Test" through Service Diagnostics to verify repair.

GAS DRYER (U.S. and Canadian Installations):

1. Unplug dryer or disconnect power.
2. Remove the cover plate from the top right corner of the back of the dryer. See figure 12.
3. Check that the power cord is firmly connected to the dryer's wire harness. See figure 14.

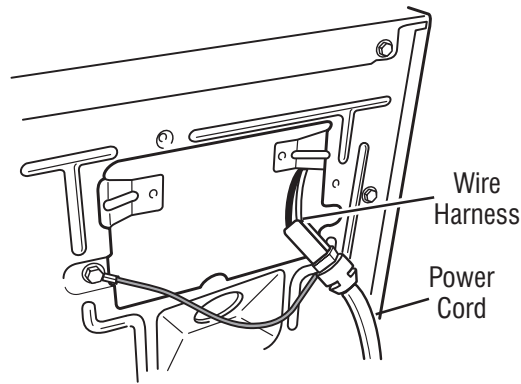


Figure 14 - Power cord-to-wire harness connection for gas dryer.

4. Access the machine electronics without disconnecting any wiring (See Dryer Disassembly Instructions, page 23).

5. With an ohmmeter, check for continuity between the neutral (N) terminal of the plug and P8-3 (white wire) on the CCU. The left-hand side of figure 15 shows the position of the neutral terminal (N) on the power cord plug. Also see figure 11, page 9.

- If there is continuity, go to step 6.
- If there is no continuity, disconnect the white wire of the main harness from the power cord at the location illustrated in figure 14. Test the continuity of the power cord neutral wire as illustrated in figure 15. If an open circuit is found, replace the power cord. Otherwise, go to step 6.

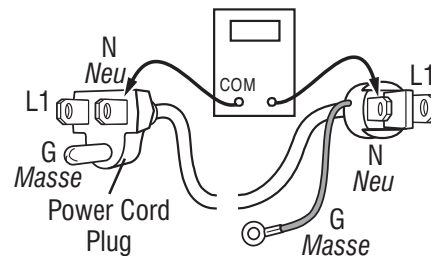


Figure 15 - Power cord terminals, gas dryer.

6. In a similar way, with an ohmmeter, check for continuity between the L1 terminal of the plug and P9-2 (black wire) on the CCU.

- If there is continuity, go to step 8.
- If there is no continuity, check the continuity of the power cord in a similar way to that illustrated in figure 15, but for power cord's L1 wire.
- If an open circuit is found, replace the power cord. Otherwise, replace the main harness.

7. If the power cord passes the continuity check, replace the main harness.
8. Visually check that ALL connectors are fully inserted into their circuit boards.
9. Reassemble all parts and panels.
10. Plug in dryer or reconnect power.
11. Perform the "Quick Diagnostic Test" through Service Diagnostics to verify repair.

TEST #3: Motor Circuit

- ✓ Quick Check: Motor activation and related attributes are provided in "Service Diagnostics → Component Activation → Motor".

This test will check the wiring to the motor and the motor itself. The following items are part of this motor system:

Part of Motor System	Electric Dryer	Gas Dryer
Drum belt	✓	✓
Door switch	✓	✓
Harness/connection	✓	✓
Thermal fuse	✓	no
Drive motor	✓	✓
Belt switch	✓	✓
Centrifugal switch	✓	✓
Machine control electronics	✓	✓

NOTE: Refer to strip circuit on page 22 to diagnose drive motor.

1. Unplug dryer or disconnect power.
2. Remove top panel to access the machine electronics.
3. Check for loose, worn, or damaged drum belt—repair as necessary.
4. Door Switch problems can be uncovered by following procedure under TEST #7: Door Switch, page 20; however, if this was not done, the following can be performed without applying power to the dryer. Connect an ohmmeter across CCU P8-3 (neutral, white wire) and P8-4 (door, tan wire).
 - With the door properly closed, the ohmmeter should indicate a closed circuit (0–2 Ω).
 - If not, check harnesses and connections between CCU and door switch. If good, replace the door switch assembly.
5. Motor Circuit Check - Access the CCU and measure the resistance across P8-4 and P9-1.

- If resistance across P8-4 and P9-1 is in the range of 1 to 6 Ω, the motor circuit is acceptable. Replace the CCU.
 - Otherwise, continue to step 6.
6. Check the wiring and components in the path between these measurement points by referring to the appropriate wiring diagram (gas or electric) on pages 25 or 26. **NOTE:** To access motor system components, refer to Dryer Disassembly Instructions, page 23.

ELECTRIC DRYER ONLY: Check the thermal fuse. See TEST #4b: Thermal Fuse, page 17.

ALL DRYERS: Continue with step 7 below to test the remaining components in the motor circuit.

7. Check the drive motor and belt switch. Slowly remove the drum belt from the spring-loaded belt switch pulley, gently letting the belt switch pulley down. See figure 16.

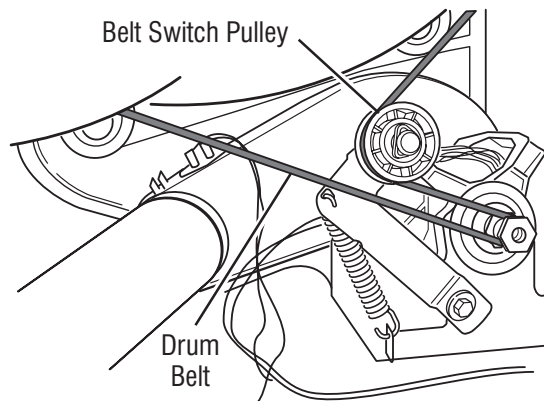


Figure 16 - Slowly remove drum belt.

8. Remove the white connector from the drive motor switch. See figure 17.

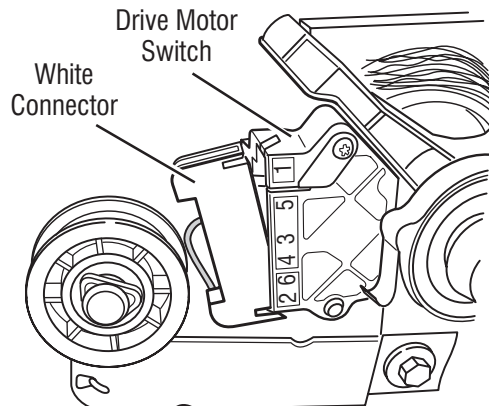


Figure 17 - Remove white connector.

9. Remove the bare copper wire terminal from pin 5 of black drive motor switch. See figure 18.

Main Winding:
Lt. Blue Wire in Back
and Bare Copper Wire

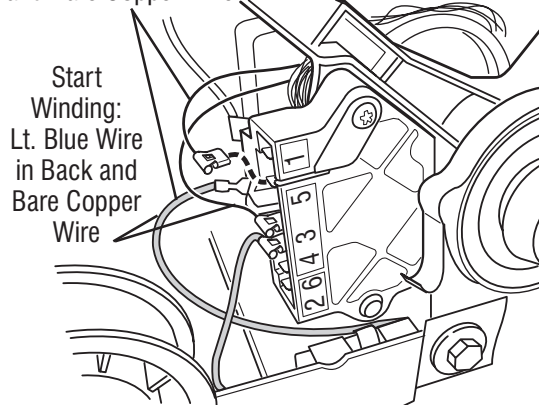


Figure 18 - Main and start winding measure points.

10. Using figure 18 and the strip circuit on page 22, check for the resistance values of the motor's Main and Start winding coils as shown in the following table.

NOTE: Main and Start winding coils must be checked at the motor.

Winding	Resistance in ohms	Contact Points of Measurement
MAIN	3.3–3.6	Lt. blue wire in back at pin 4 and bare copper wire terminal removed from pin 5 of black drive motor switch
START	2.7–3.0	Lt. blue wire in back at pin 4 and bare copper wire terminal on pin 3 of black drive motor switch

- If the resistance at the motor is correct, there is an open circuit between the motor and CCU. Go to step 11 to check for belt switch problem.
- If the Start winding resistance is much greater than 3 ohms, replace the motor.

11. Check the belt switch by measuring resistance between the two light blue wires, as shown in figure 19, while pushing up the belt switch pulley.

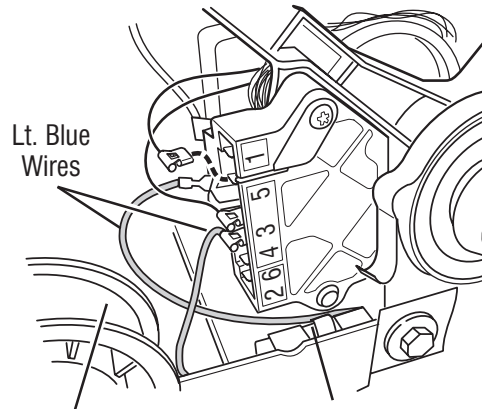


Figure 19 - Checking the belt switch.

- If the resistance reading goes from open to a few ohms as pulley arm closes the switch, belt switch is good. If not, replace the belt switch.
- If belt switch is good and there is still an open circuit, check and repair the main wiring harness.

12. Reassemble all parts and panels.

13. Plug in dryer or reconnect power.

14. Perform the "Quick Diagnostic Test" through Service Diagnostics to verify repair.

TEST #4: Heat System

- ✓ Quick Check: Heater activation and related component attributes are provided in "Service Diagnostics → Component Activation → Heater 1, Heater 2, and Heater 1 + 2" for electric, or "Heat On" for gas.

This test is performed when either of the following situations occurs:

- ✓ Dryer does not heat
- ✓ Heat will not shut off

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This test checks the components making up the heating circuit. The following items are part of this system:

Part of Heating System	Electric Dryer	Gas Dryer
Harness/connection	✓	✓
Heater relay	✓	✓
Thermal cut-off	✓	✓
Thermal fuse	no	✓
High limit thermostat	✓	✓
Heat element assembly	✓	no
Gas valve assembly	no	✓
Centrifugal switch	✓	✓
Outlet thermistor	✓	✓
Inlet thermistor	✓	✓
Machine control electronics	✓	✓
Console electronics and housing assembly	✓	✓
Gas supply	no	✓

NOTE: On the gas dryer, the inlet thermistor is located below the CCU bracket at the drum inlet vent. Refer to strip circuit on page 22 to diagnose heater system.

Dryer does not heat:

Locate the components using figures 20a and 20b. To access heater system components, see Dryer Disassembly Instructions, page 23.

ELECTRIC DRYER ONLY:

- ✓ Quick Check: Perform the “CCU Line Voltage 1” check under Component Activation. If L1 is present, the thermal cutoff is functional.
- ✓ Quick Check: Perform the “CCU Line Voltage 2” check under Component Activation. If L2 is present, the centrifugal switch, high limit thermostat, and the side of the heater connected to heater relay 1 are functional.

1. Unplug dryer or disconnect power.
2. Remove top panel to access the machine electronics.
3. Check Heaters—on the CCU, use an ohmmeter to measure the resistance between the violet wire terminal on heater relay #1 and the violet wire terminal on heater relay #2.
 - If the resistance is $\leq 50 \Omega$, go to step 5.
 - If an open circuit is detected, go to step 4.

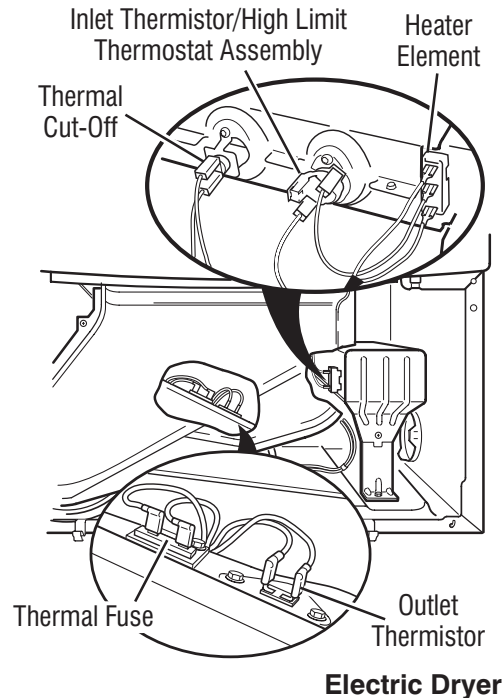


Figure 20a - Thermal components, electric dryer, viewed from front.

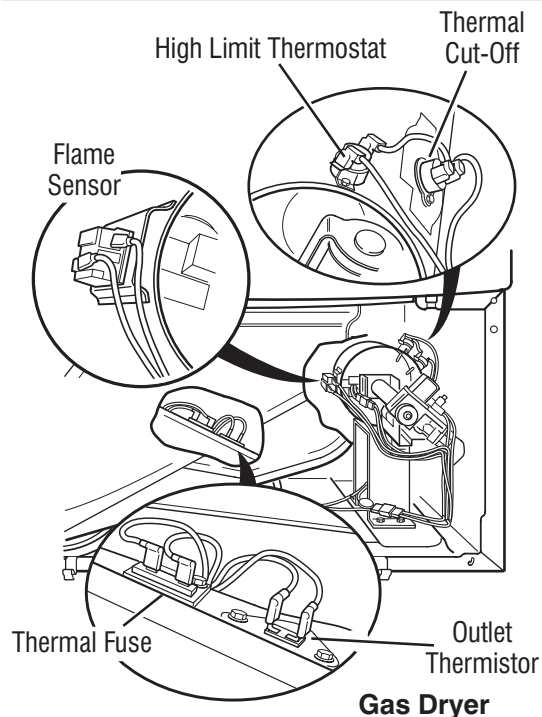


Figure 20b - Thermal components, gas dryer, viewed from front.

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4. Visually check the wire connections between each relay and their respective heaters. If the connections look good, check for continuity across each heater (violet wire to center red wire). Refer to strip circuit on page 22.

➤ Replace the heater if it is electrically open.

5. Check Thermal Cut-off—on the CCU, use an ohmmeter to measure continuity between P9-2 (L1) and the black wire terminal on heater relay #1. Then, measure continuity between P9-2 (L1) and the black terminal on heater relay #2.

➤ If there is continuity, go to step 7.

➤ If an open circuit is detected, go to step 6.

6. Visually check the wire connections between each relay (black wire) and the thermal cut-off. If the connections look good, check for continuity across the thermal cut-off.

➤ Replace the thermal cut-off if it is electrically open.

7. Check High Limit Thermostat—visually check the wire connections from the heaters and centrifugal switch to the high limit thermostat. If the connections look good, check for continuity across the high limit thermostat.

➤ Replace the high limit thermostat if it is electrically open.

8. If no open circuit is detected, remove P14 from the CCU and measure the outlet thermistor resistance between P14-3 and P14-6 at the connector.

➤ If 5–15 k ohms are measured, go to step 9.

➤ If the resistance is less than 1 k ohm, replace the outlet thermistor.

9. If the preceding steps did not correct the problem and L1 and L2 were both detected, replace the CCU. If L2 was not detected, suspect the centrifugal switch before replacing the CCU.

10. Reassemble all parts and panels.

11. Plug in dryer or reconnect power.

12. Perform the “Quick Diagnostic Test” through Service Diagnostics to verify repair.

GAS DRYER ONLY:

1. Verify that the gas supply to the dryer is turned on.

2. Unplug dryer or disconnect power.

3. Perform TEST #4b: Thermal Fuse on page 17. If the thermal fuse is OK, go to step 4.

4. Perform TEST #4c: Thermal Cut-Off on page 17. If the thermal cut-off is OK, go to step 5.

5. Locate the high limit thermostat (see figures 20a & 20b, page 14). Measure the continuity through it by connecting the meter probes to the red and black wire terminals.

➤ If there is an open circuit, replace both the high limit thermostat and the thermal cut-off.

➤ Otherwise, go to step 6.

6. Perform TEST #4d: Gas Valve on page 17. If the gas valve is OK, go to step 7.

7. If the preceding steps did not correct the problem, suspect the centrifugal switch before replacing the CCU.

8. Reassemble all parts and panels.

9. Plug in dryer or reconnect power.

10. Perform the “Quick Diagnostic Test” through Service Diagnostics to verify repair.

Heat will not shut off:

ALL DRYERS:

1. Unplug dryer or disconnect power.

2. Remove top panel to access the machine electronics.

3. Remove connector **P14** from the CCU and measure the resistance between P14-3 and P14-6 at the connector.

➤ If 5–15 k ohms are measured, the outlet thermistor is OK.

➤ If the resistance is less than 1 k, replace the thermistor; if open, repair as necessary.

4. Check heater coil(s) for a short to ground (usually inside the heater box). Repair or replace if necessary.

5. Plug in dryer or reconnect power.

6. Run an “AIR” only timed dry cycle (no heat). Check heater relay output(s) on CCU. With a voltmeter set to **AC**, measure the voltage across terminals 1 & 2 for heater relay #1. If equipped, repeat procedure with heater relay #2.

➤ If voltage is present (~240VAC for electric, ~120VAC for gas), the relay is open and working normally.

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- If little or no voltage is present, the relay is closed and heater is activated. Replace the CCU.
- 7. Unplug dryer or disconnect power.
- 8. Reassemble all parts and panels.
- 9. Plug in dryer or reconnect power.
- 10. Perform the "Quick Diagnostic Test" through Service Diagnostics to verify repair.

TEST #4a: Thermistors

NOTE: Refer to strip circuit on page 22 to diagnose outlet and inlet temperature thermistors.

Outlet (Exhaust) Thermistor

The CCU monitors the exhaust temperature using the outlet thermistor, and cycles the heater relay on and off to maintain the desired temperature. **NOTE:** Begin with an empty dryer and a clean lint screen.

- ✓ Quick Check: Exhaust thermistor readings in °F, °C, and resistance are provided in "Service Diagnostics → Component Activation → Exhaust Thermistor."

1. Unplug dryer or disconnect power.
2. Remove top panel to access the machine electronics.
3. Remove connector **P14** from the CCU and measure the resistance between P14-3 and P14-6 at the connector. The following table gives temperatures and their associated resistance values.

NOTE: All thermistor resistance measurements must be made while dryer is unplugged and connector removed from CCU.

OUTLET THERMISTOR RESISTANCE			
TEMP. °F (°C)	RES. RANGE k ohms	TEMP. °F (°C)	RES. RANGE k ohms
50° (10°)	19.0–22.0	80° (27°)	8.5–10.5
60° (16°)	14.8–16.8	90° (32°)	6.8–8.8
70° (21°)	11.5–13.5	100° (38°)	5.0–7.0

- If the resistance is OK, the outlet thermistor is good. Proceed to step 4.
 - If the thermistor resistance does not agree with the table, replace the outlet thermistor.
4. Check P14-3 and P14-6 to dryer cabinet ground. If either pin indicates continuity to ground (short), replace wiring harness; otherwise, proceed to step 5.

5. If the preceding steps did not correct the problem, replace the CCU.

Temperature Levels Incorrect – If no error code is displayed and the connections to the thermistor are good, check the exhaust temperature value at any or all of the temperature levels in question, using the **Timed Dry** cycle. **IMPORTANT:** Be sure EcoBoost (if available) is turned OFF before testing.

1. Remove load from dryer and disconnect external vent.
2. Plug in dryer or reconnect power.
3. Run a **TIMED DRY** cycle of at least 2 minutes in duration. Select High, Medium High, Medium, Low, or Extra Low. Turn off EcoBoost.
4. Using a calibrated temperature probe, take a temperature measurement in the center of the exhaust outlet. The correct exhaust temperatures are as follows:

EXHAUST TEMPERATURES		
TEMPERATURE SETTING	HEAT TURNS OFF* °F (°C)	HEAT TURNS ON °F (°C)
High	155° ± 5° (68° ± 3°)	10–15° (6–8°) below the heat turn off temperature
Medium High	150° ± 5° (65° ± 3°)	
Medium	140° ± 5° (60° ± 3°)	
Low	125° ± 5° (52° ± 3°)	
Extra Low	105° ± 5° (41° ± 3°)	

- If the temperature is not reached within ~7 minutes, check voltage level and vent blockage, and then retest.
- If the temperature probe does not agree with temperature setting, replace the outlet thermistor.
- If the temperature probe confirms the temperature setting, retest at a different temperature setting.

5. If the preceding steps did not correct the problem, replace the CCU.

Inlet Thermistor

The CCU monitors the inlet temperature using the inlet thermistor. The inlet thermistor (along with the outlet thermistor) is used to detect air flow, and assists in calculating load size.

NOTE: On the electric dryer, the inlet thermistor is part of the high thermostat assembly (see figures 20a & 20b). On the gas dryer, the inlet thermistor is located below the CCU bracket at the drum inlet duct (see figure 28, page 27).

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- ✓ Quick Check: Inlet thermistor readings in °F, °C, and resistance are provided in "Service Diagnostics → Component Activation → Inlet Thermistor."

1. Unplug dryer or disconnect power.
2. Remove top panel to access the machine electronics.
3. Remove connector **P14** from the CCU and measure the resistance between P14-1 and P14-2 at the connector. The following tables (electric & gas) give temperatures and their associated resistance values.

NOTE: All thermistor resistance measurements must be made while dryer is unplugged and connector removed from CCU.

- If the resistance is OK, the inlet thermistor is good. Proceed to step 4.
- If the thermistor resistance does not agree with the table, replace the inlet thermistor.

ELECT - INLET THERMISTOR RESISTANCE			
TEMP. °F (°C)	RES. RANGE k ohms	TEMP. °F (°C)	RES. RANGE k ohms
68° (20°)	61.2–63.7	131° (55°)	14.5–15.3
77° (25°)	49.0–51.0	140° (60°)	12.1–12.8
86° (30°)	39.5–41.1	149° (65°)	10.2–10.7
95° (35°)	32.0–33.3	158° (70°)	8.5–9.0
104° (40°)	26.1–27.2	167° (75°)	7.2–7.6
113° (45°)	21.4–22.3	176° (80°)	6.1–6.5
122° (50°)	17.6–18.5		

GAS - INLET THERMISTOR RESISTANCE			
TEMP. °F (°C)	RES. RANGE k ohms	TEMP. °F (°C)	RES. RANGE k ohms
68° (20°)	57.5–67.6	131° (55°)	14.1–15.6
77° (25°)	46.1–53.8	140° (60°)	11.8–12.9
86° (30°)	37.4–43.1	149° (65°)	9.9–10.8
95° (35°)	30.4–34.7	158° (70°)	8.4–9.0
104° (40°)	24.9–28.2	167° (75°)	7.1–7.6
113° (45°)	20.5–23.0	176° (80°)	6.0–6.4
122° (50°)	16.9–18.9		

4. Check P14-1 and P14-2 to dryer cabinet ground. If either pin indicates continuity to ground (short), replace wiring harness; otherwise, proceed to step 5.
5. If the preceding steps did not correct the problem, replace the CCU.

TEST #4b: Thermal Fuse

ELECTRIC DRYER: The thermal fuse is wired in series with the dryer drive motor.

GAS DRYER: The thermal fuse is wired in series with the dryer gas valve.

ALL DRYERS:

1. Unplug dryer or disconnect power.
2. Access the thermal fuse by removing the front panel. See Dryer Disassembly Instructions, page 23.
3. Using an ohmmeter, check the continuity across the thermal fuse.
 - If the ohmmeter indicates an open circuit, replace the thermal fuse.

TEST #4c: Thermal Cut-Off

If the dryer does not produce heat, check the status of the thermal cut-off.

1. Unplug dryer or disconnect power.
2. Access the thermal cut-off by removing the front panel. See Dryer Disassembly Instructions, page 23.
3. Using an ohmmeter, check the continuity across the thermal cut-off. See figures 20a and 20b, page 14, for location.
4. If the ohmmeter indicates an open circuit, perform the following:

ALL DRYERS: Replace both the thermal cut-off and high limit thermostat. In addition, check for blocked or improper exhaust system, and, on electric dryers, for heat element malfunction.

TEST #4d: Gas Valve (Gas Dryer)

- ✓ Quick Check: Heater activation and related component attributes are provided in "Service Diagnostics → Component Activation → Heat On."

1. Unplug dryer or disconnect power.
2. Access the gas valve by removing the back panel. See Dryer Disassembly Instructions, page 23.
3. Use an ohmmeter to determine if a gas valve coil has malfunctioned. Remove harness plugs. Measure resistance across the terminals (see figure 21, page 18). Readings should match those shown in the following chart; if not, replace coils.

GAS VALVE RESISTANCE	
Terminals	Resistance in ohms
1 to 2	1400 ± 70
1 to 3	570 ± 28.5
4 to 5	1300 ± 65

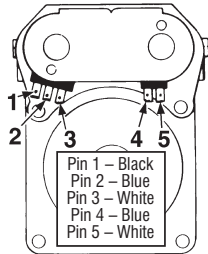


Figure 21 - Measuring gas valve resistance.

4. Disconnect the ignitor plug from the burner. Using an ohmmeter, measure the resistance across the ignitor's 2-pin connector. Resistance should be 50-500 Ω.

- If resistance readings are outside the range or open, replace the ignitor.
- If resistance readings are within range, reconnect the ignitor plug and continue to step 5.

5. Disconnect the wires going to the flame sensor terminals. Using an ohmmeter, measure across the two sensor terminals for continuity.

- If there is continuity, reconnect the sensor wires and continue to step 6.
- If the reading is open, the flame sensor needs replacing.

6. Reassemble back panel before reconnecting power.

7. Plug in dryer or reconnect power.

8. Run a high-temp TIMED DRY cycle of at least 2 minutes in duration.

9. Watch the ignitor for a couple of minutes through the "peek window" in the side. If the ignitor stays red hot and the gas does not come out and ignite, the flame sensor needs replacing.

NOTE: If ignitor does not come on, line voltage may not be present at the gas burner. The motor centrifugal switch may be suspect.

IMPORTANT: To avoid damage to the gas burner wire harness, ensure the harness is routed exactly as it was prior to service.

10. Unplug dryer or disconnect power.

11. Reassemble all parts and panels.

12. Plug in dryer or reconnect power.

13. Perform the "Quick Diagnostic Test" through Service Diagnostics to verify repair.

TEST #5: Moisture Sensor

This test is performed when an automatic cycle stops too soon, or runs much longer than expected.

NOTE: Dryer will shut down automatically after 2½ hours.

- ✓ Quick Check: Moisture Sensor status (open/closed) is provided in "Service Diagnostics → Component Activation → Moisture Sensor."

The following items are part of this system:

Part of Moisture System	Electric Dryer	Gas Dryer
Harness/connection	✓	✓
Metal sensor strips	✓	✓
Machine control electronics	✓	✓

NOTE: Refer to strip circuit on page 22 to diagnose moisture sensor.

1. Activate Service Diagnostics and then select the Component Activation → Moisture Sensor. See procedures on page 5.

2. Open the door. Using a wet cloth or one finger, jointly touch both sensor strips.

- If the sensor status changes from open to closed, the sensor passed the test. Go to step 9.
- If the sensor status remains open, or was closed before touching both moisture strips, continue with step 3.

NOTE: Overdrying may be caused by a short circuit in the sensor system.

3. Unplug dryer or disconnect power.

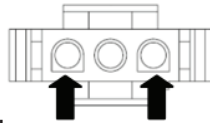
4. Remove top panel to access the machine electronics.

5. Access the moisture sensor wires by removing the console and front panel (see page 23). Disconnect the 3-wire moisture sensor connector located below the door opening between the front panel and bulkhead.

6. Access the CCU and remove connector P13 from the circuit board. Check the wire harness for continuity between P13 and the moisture sensor connector.

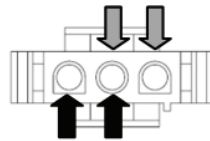
- If there is continuity, go to step 7.
- If there is no continuity, or if a short is detected, replace the main harness.

7. Measure the resistance across the outermost contacts of the connector that includes the two MOVs.



- If a small resistance is measured, clean the two metal moisture strips inside the drum. If a small resistance is measured after cleaning, replace sensor harness.
- If a small resistance is not measured, go to step 8.

8. Measure the resistance across each of the outermost contacts and the center terminal (ground connection).



- If a resistance less than infinity is measured, replace the sensor harness.
9. If the moisture sensor diagnostic test passes, check the outlet thermistor: TEST #4a, page 16.
- If the problem persists after replacing the moisture sensor and thermistor, consider adjusting the dryness level (see TEST #5a: Adjusting Customer-Focused Dryness Level).
10. If the preceding steps did not correct the problem, replace the CCU.
- a. Unplug dryer or disconnect power.
 - b. Replace the CCU.
 - c. Reassemble all parts and panels.
 - d. Perform the "Quick Diagnostic Test" through Service Diagnostics to verify repair.

TEST #5a: Adjusting Customer-Focused Dryness Level

NOTE: If the customer complains about the clothes being damp and the moisture sensor passes TEST #5: Moisture Sensor, step 3, the total dry time can be lengthened by changing the factory preset dryness setting.

1. Turn the cycle selector knob to an automatic cycle.
2. From the Automatic Cycle Main Screen, press and hold the Dryness Level button (~3 sec.) until the Customer-Focused Dryness Level screen appears.
3. Select the desired dryness setting: "Factory Preset" (standard auto cycle), "15% More", "30% More", "15% Less", or "30% Less".

The following screen will confirm your selection. The selection will be stored on the CCU and will be retained—even after a power loss.

NOTE: If dryness setting is not selected within 6 seconds, it will time out and exit to the Main Screen.

TEST #6: Buttons and Indicators

This test is performed when any of the following situations occurs during the "Service Diagnostics → Component Activation → UI Component Test." (See page 5.)

- ✓ **None of the indicators or display turn on**
- ✓ **Some buttons do not light**
- ✓ **No beep sound is heard**

NOTE: The User Interface, which includes the Console, Modifier, and Center PCBs, must be diagnosed and replaced as a single component.

None of the indicators or display turn on:

1. Unplug dryer or disconnect power.
2. Remove top panel to access the machine electronics.
3. Visually check that ALL connectors are fully inserted into the CCU.
4. Visually check that ALL connectors are fully inserted into the UI.
5. Visually check that the UI and housing assembly is properly inserted into the front console.
6. If all visual checks pass, perform TEST #1: CCU Power Check, page 9, to verify supply voltages.
 - If supply voltages are present, replace the user interface and housing assembly.
 - If supply voltages are not present, replace the CCU.
7. Reassemble all parts and panels.
8. Plug in dryer or reconnect power.
9. Perform the "UI Component Test" to verify repair.

Some buttons do not light:

1. Unplug dryer or disconnect power.
2. Remove top panel to access the machine electronics.

3. Visually check that the UI and housing assembly is properly inserted into the front console.
4. If visual check passes, replace the UI and housing assembly.
5. Reassemble all parts and panels.
6. Plug in dryer or reconnect power.
7. Perform the "UI Component Test" to verify repair.

No beep sound is heard:

1. Check volume levels under System Settings → End of Cycle and Button Sound. Increase volume if needed.
2. Unplug dryer or disconnect power.
3. Remove top panel to access the machine electronics.
4. Visually check that ALL connectors are fully inserted into the CCU.
5. Visually check that ALL connectors are fully inserted into the UI.
6. If all visual checks pass, replace the UI and housing assembly.
7. Reassemble all parts and panels.
8. Plug in dryer or reconnect power.
9. Perform the "UI Component Test" to verify repair.

TEST #7: Door Switch

- ✓ Quick Check: Door Switch status (open/closed) is provided in "Service Diagnostics → Component Activation → Door Status."

Functionality is verified when the door status toggles between Open to Closed. If the preceding condition is not met:

1. Unplug dryer or disconnect power.
2. Remove top panel to access the machine electronics.
3. Check that the wires between the door switch and CCU are connected. (Refer to wiring diagrams on pages 25 and 26.)
 - If the connections are good, replace the wire and door switch assembly and retest.
 - If wire and door switch assembly have been replaced and dryer still does not start, replace the CCU.
4. Reassemble all parts and panels.

5. Plug in dryer or reconnect power.
6. Perform the "Quick Check" above to verify repair.

TEST #8: Drum Light

- ✓ Quick Check: Drum Light status (on/off) is provided in "Service Diagnostics → Component Activation → Drum Light."

Functionality is verified when the drum light turns on and off. If the preceding condition is not met:

1. Press and hold the EcoBoost/Drum Light button on the console until the LCD screen acknowledges the drum light is on.
 - If the LCD does not confirm the button press, replace the user interface (UI).
 - If the LCD confirms the button press, go to step 2.
2. Unplug dryer or disconnect power.
3. Remove the top panel to access CCU and UI.
4. Verify the drum LED connector **P13** is securely connected to the UI.
5. Check harness and inline connections between the drum LED and UI.
 - If the harness and connections are good, go to step 6.
 - If not, repair or replace as needed.
6. Unplug the drum LED connector **P13** from the UI.
7. Plug in dryer or reconnect power.
8. Press and hold the EcoBoost/Drum Light button on the console to activate the LED circuit.
9. With a multimeter set to milliamps, measure the current across UI connector P13, pins 1 and 3. If the drum LED driver is working properly, there should be a measurement of 150–370 mA.
 - If the current is present, replace the drum LED.
 - If the current is not present, replace the UI.
10. Unplug dryer or disconnect power.
11. Reassemble all parts and panels.

TEST #9: Myst Valve

- ✓ Quick Check: Myst Valve status (on/off) is provided in "Service Diagnostics → Component Activation → Myst Valve."

Functionality is verified when the valve turns on and off and water is being sprayed into the drum (see figure 22).

NOTE: Refer to strip circuit on page 22 to diagnose myst valve.

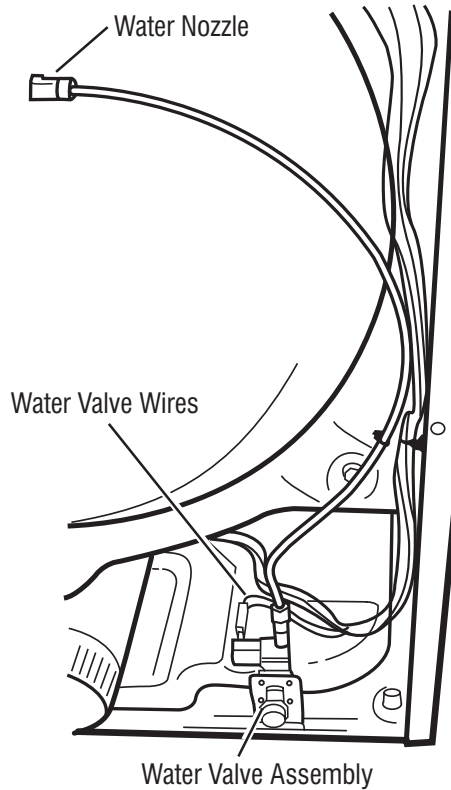


Figure 22 - Water System Components.

If no water is sprayed in the drum:

1. Verify that water is hooked up and turned on.
2. Unplug dryer or disconnect power.
3. Remove top panel to access the machine electronics.
4. Verify that the red wire coming from the water valve is connected to the CCU, P8-1. Refer to wiring diagrams on pages 25 and 26.
5. Check Myst Valve & Harness—using an ohmmeter, measure the resistance between the CCU P8-1 (red wire) and P9-2 (black wire).
 - If the resistance is 510–590 Ω , go to step 6; if not, replace the myst valve.
 - If an open circuit is detected, go to step 7.

6. Inside the drum, unscrew and replace the myst nozzle using a 7/16" wrench or socket. Retest myst valve.

➤ If water does not dispense, go to step 7.

7. Access the myst valve by removing back panel; see page 23.

➤ Check that the hose and wires are connected to the myst valve assembly (see figure 22).

➤ Check that the myst valve assembly hose is connected to the nozzle.

8. If everything is hooked up and the water still does not dispense:

➤ Unplug dryer or disconnect power.

➤ Replace the valve assembly and retest.

9. If the preceding steps did not correct the problem, replace the CCU.

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STRIP CIRCUITS

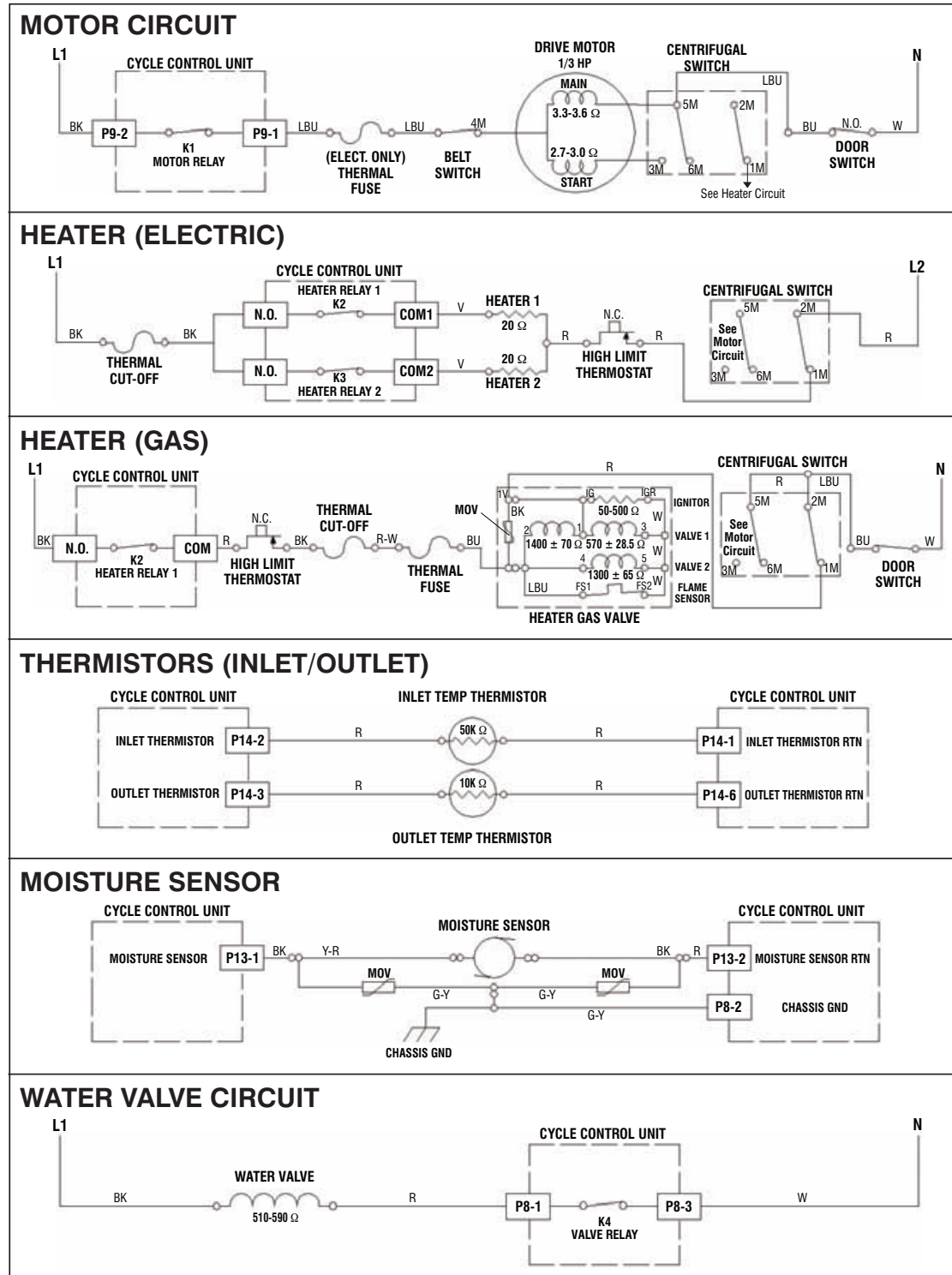
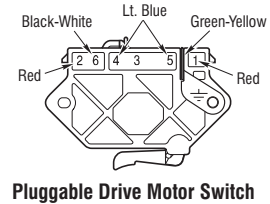
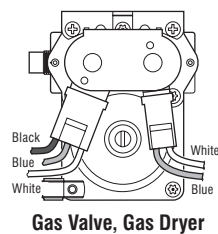
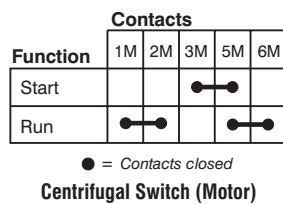


Figure 23 - Strip Circuits.



DRYER DISASSEMBLY INSTRUCTIONS

Remove Top Panel

1. Unplug dryer or disconnect power.
2. Remove the three rear screws securing the top panel and extension bracket to the back panel.
3. Slide the top panel/extension bracket back to remove.
4. Reverse procedure to reassemble.

Remove Rear Panel (Figure 24)

1. Perform "Remove Top Panel".
 2. **ELECTRIC:** Remove the cover plate and disconnect the power cord from the terminal block and ground screw.
 3. **GAS:** Remove the cover plate and disconnect the AC power cord from the ground screw.
 3. Remove the two screws securing the terminal block to rear panel.
 4. Remove the 13 screws securing the rear panel to the cabinet. Remove rear panel from dryer.
- NOTE:** The spring clip has been replaced by a lint duct collar that is secured to the rear panel. Care must be taken when sliding the panel off the exhaust outlet because of the tight fit.
5. Reverse procedure to reassemble.



Figure 24 - Remove rear panel.

Remove Console, Front Panel, Front Bulkhead, and Drum

NOTE: Sections A, B, C, and D must be performed in order.

A: Remove Console (Figure 25)

1. Perform "Remove Top Panel".
2. Disconnect the drum light harness at the inline connector as shown in figure 25.
3. Disconnect the CCU to UI harness at the inline connection as shown in figure 25.
4. Remove the two top screws (closest to the front) securing console to mounting bracket.
5. Slide the console up and off the mounting bracket while feeding the CCU to UI and drum light harnesses through the opening in the bracket. Remove console and harnesses from dryer.

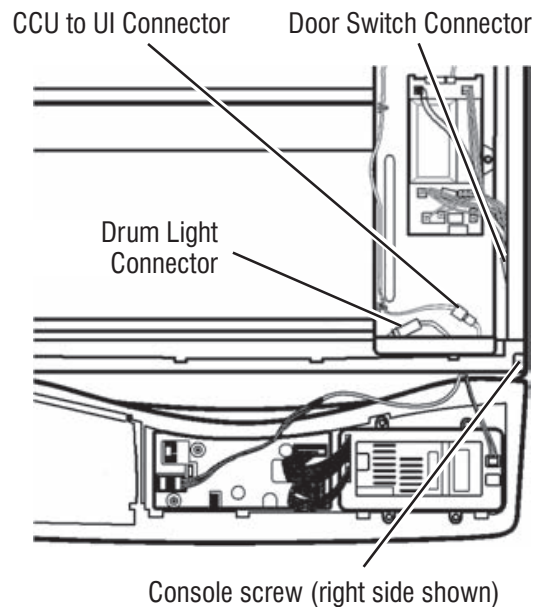


Figure 25 - Remove console.

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B: Remove Front Panel

(Refer to Figure 26, unless otherwise noted.)

6. Disconnect door switch connector (See Figure 25, page 23).

7. Remove front panel by removing nine screws—four from the bottom, three from the top, and two below the dryer door opening.

Console Mounting Bracket

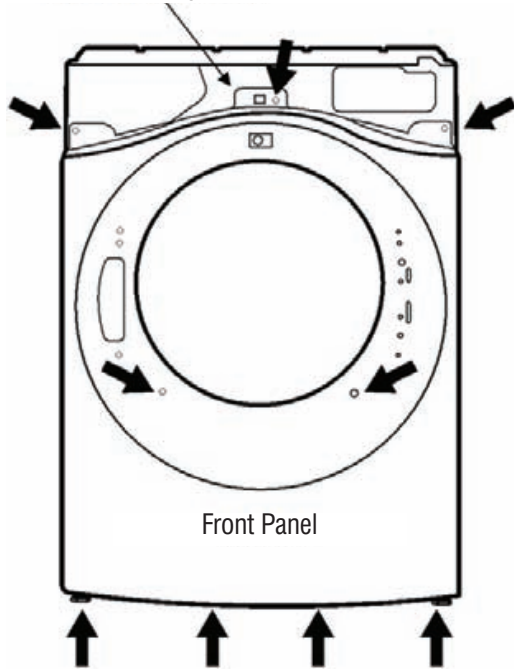


Figure 26 - Remove front panel.

C: Remove Front Bulkhead

(Refer to Figure 27, unless otherwise noted.)

8. Remove lint screen.

9. Unclip main wire harness (2 places).

10. Disconnect drum light connector (See Figure 25, page 23).

11. Disconnect moisture sensor at plug below door opening.

12. Remove console mounting bracket by removing six screws—four from the top and two from the front (bold arrows).

13. Remove lint duct cover by removing three screws—one screw and clamp assembly at bottom of cover and two screws securing cover to bulkhead.

14. Remove the four screws securing the bulkhead to the cabinet. Remove bulkhead.

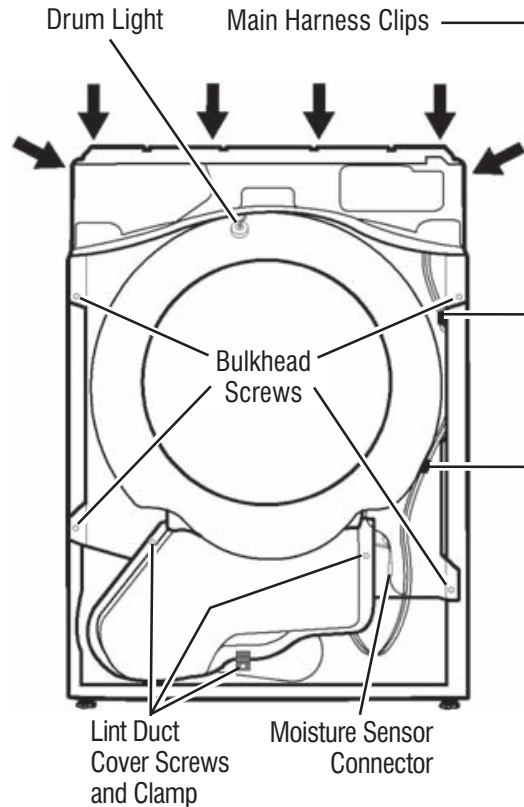


Figure 27 - Remove front bulkhead.

D: Remove Drum

15. Remove drive belt from belt switch pulley.

16. Pull drum from cabinet.

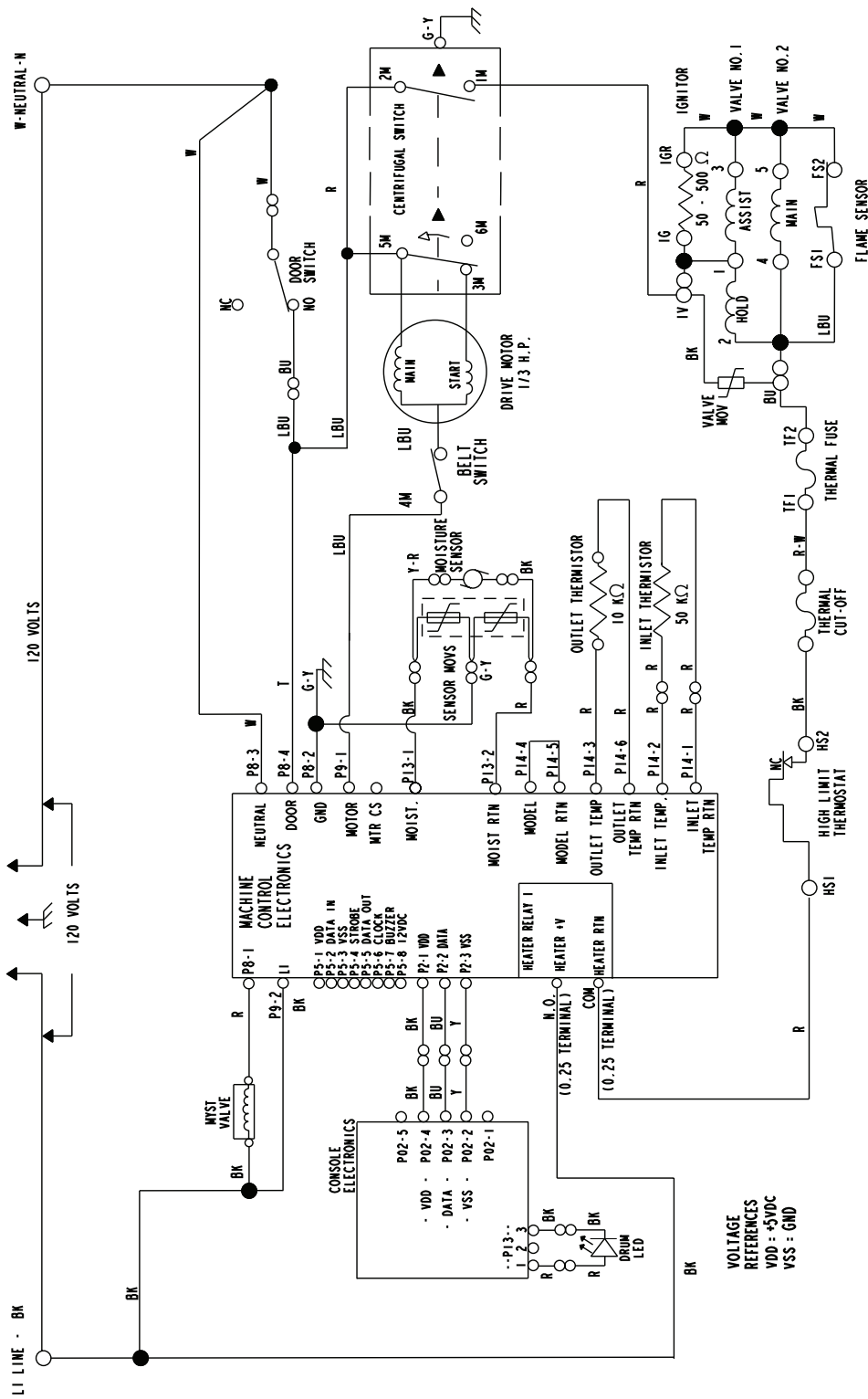
17. Reverse procedure to reassemble.

DO NOT REMOVE OR DESTROY



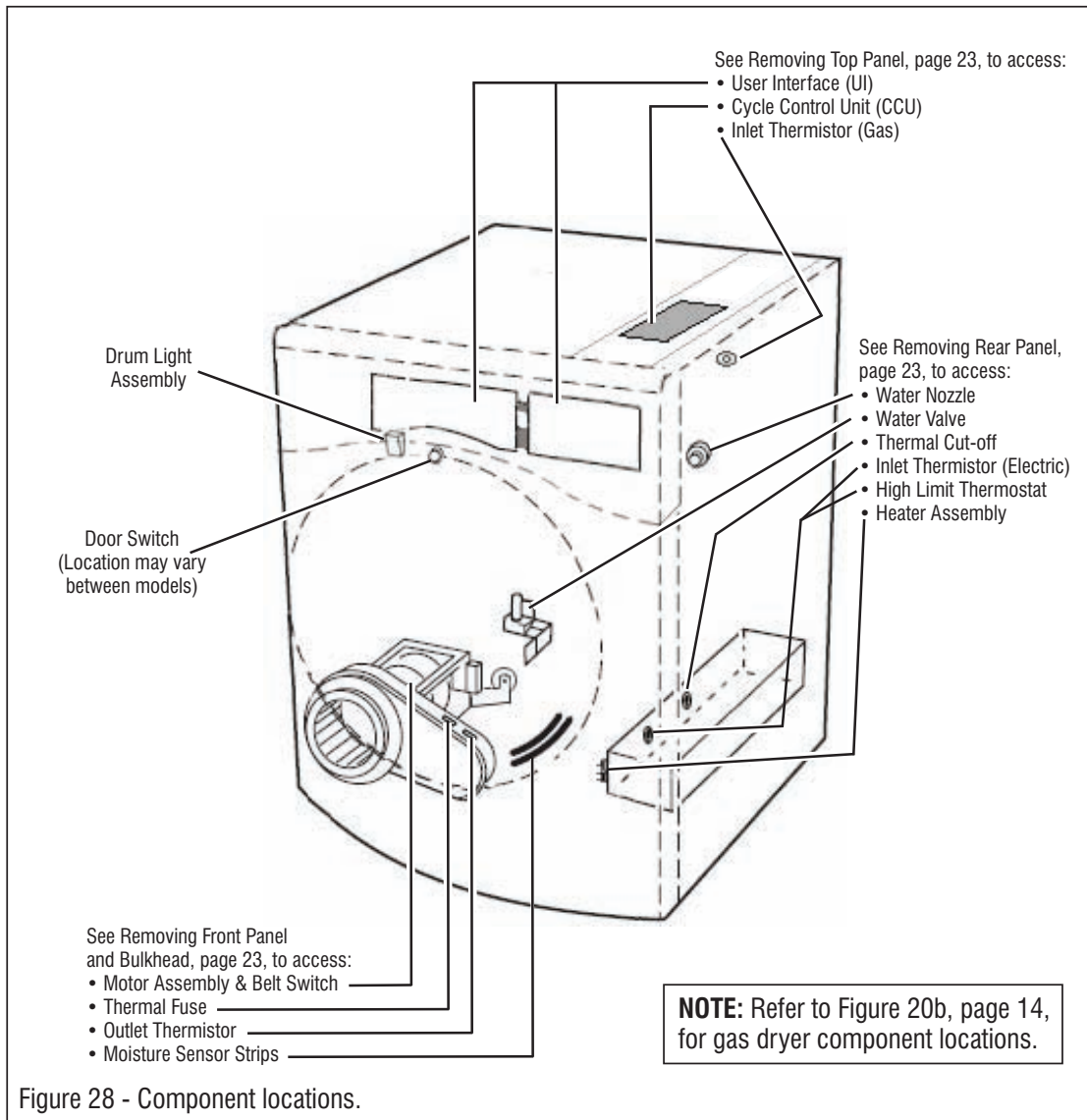
WHIRLPOOL GAS DRYER WIRING DIAGRAM

IMPORTANT: Electrostatic discharge may cause damage to machine control electronics. See page 1 for ESD information.



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COMPONENT LOCATIONS



SPECIFICATIONS

DRYER SPECIFICATIONS	
Voltage:	240 VAC (200-260) Elect. Dryer, 2-phase, "optimized" 208 VAC (176-229) Elect. Dryer, 3-phase, "less optimized" 120 VAC (100-130) Gas Dryer
Amps:	(ELECT) 30 Amp Service (GAS) 15 Amp Service
Frequency:	58 to 62 Hz (60 Hz nominal)
Water Pressure:	20-120 PSI
Operating Temperature Range:	40 to 105°F (5 to 40°C)
Dryer Height:	39 in. (99.1 cm)
Dryer Width:	27 in. (68.6 cm)
Dryer Depth:	31 in. (78.7 cm)

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NOTES